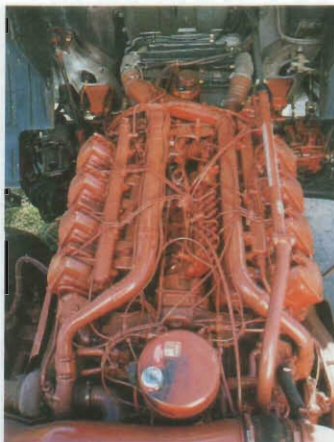




Eight-cylinder diesel truck engine

Diesel engine

A diesel engine is a four-stroke engine without spark plugs. The engine's cylinder has a piston, which rises and falls, squashing the fuel-and-air mixture in the cylinder into a tiny space. The mixture gets so hot, it explodes.

Using diesels

Diesel engines are very fuel-efficient. They are used for driving electricity generators, and in vehicles that need to keep going for long periods without refuelling, such as lorries, taxis, trains, ships, and boats. Many modern cars are also fitted with diesel engines.

Steam engine

The pistons of a steam engine are moved up and down in their cylinders by steam under high pressure. The pistons are connected to rods that turn the wheels. The steam is made outside the cylinders by heating water in a coal-fired boiler, which is why steam engines are called external combustion engines.

Using steam

Until the middle of the 1900s, most railway locomotives and ships were powered by steam engines. Steam also drove many early trucks and buses. The first steam engines were used for pumping flood water out of mines, and to work industrial machines.

Steam leaves train via a funnel.



Steam engine Firebox Pipes heat water, which turns into steam. Steam is fed into the cylinder. Piston

Solar power

Petroleum and coal are fossil fuels, formed from decayed prehistoric organisms. They are expensive to produce, and create harmful gases when they burn. Solar energy is energy from the Sun. It can be used to heat houses, run air conditioning, and to generate electricity to power lightweight vehicles.

Solar panels

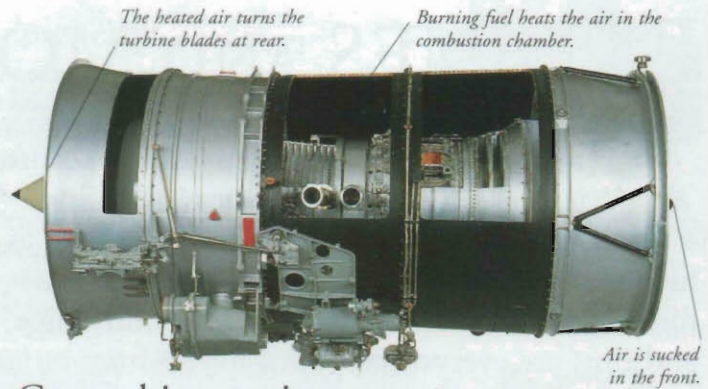
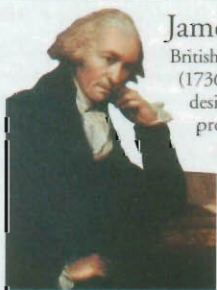
Sunlight can be turned into electricity by solar panels. These are made from many photovoltaic cells. The bigger the area of photovoltaic cells, and the brighter the sunlight, the larger the electric current the solar panel will produce.



Solar-powered car

James Watt

British engineer James Watt (1736–1819) improved the design of steam engines, and produced the first effective one in 1765. In 1774, he and Matthew Boulton began building steam engines for pumping water from mines. The unit of power, the watt, is named after him.



The heated air turns the turbine blades at rear.

Burning fuel heats the air in the combustion chamber.

Air is sucked in the front.

Gas turbine engines

In a gas turbine, burning fuel makes a stream of hot gas that spins a set of turbine blades very fast. A shaft attached to the turbine drives a compressor that sucks air into the engine so the fuel burns.

Jet engine

High-speed aircraft have a type of turbine called a turbojet or turbofan. The stream of hot air and gases created in the engine turns the turbine, then shoots out of the back of the engine, pushing the aircraft forwards.

Turboshaft engine

Some turbine engines make ship or aircraft propellers spin. The spinning turbine turns a shaft connected to the propeller. Large hovercraft have turboshaft engines to create their air cushion and to drive their propellers. Large helicopters also have turboshaft engines to turn their rotors.



SR.N4 ferry hovercraft

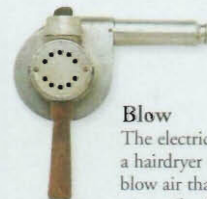
Electric motors

An electric motor produces movement from electricity. Inside it are electromagnets – wire coils that become magnets when an electric current flows through them. The electromagnets are turned on and off in sequence to pull a magnetic shaft around and around. Motors are used in household appliances.

Suck

A vacuum cleaner has a powerful electric motor that turns an air pump. The pump sucks air through the machine, where the dust is removed from it. The motor has to generate a lot of power, so it needs electricity from the mains to drive it.

Vacuum cleaner



Hairdryer

Blow

The electric motor in a hairdryer turns a fan to blow air that is heated by hot wire coils. A switch adjusts the speed of the motor. The larger the current it allows through, the stronger the magnets become, and the faster the motor spins.

Turn

Many kitchen gadgets, such as food processors, have an electric motor that moves their working parts. Gears slow the speed of the motor, so the parts turn slowly. The electricity comes either from the mains or from batteries.

Food processor



Timeline

1st century AD Hero of Alexandria, a Greek inventor, makes a novelty toy that is turned by steam.

1698 Englishman Thomas Savery (c.1650–1715) builds the first machine to provide power by using steam.

1815 British engineer George Stephenson (1781–1848) builds the first steam-powered locomotive.



Gears

1876 In Germany, Nikolaus Otto (1832–91) develops the first four-stroke petrol engine. It is a great commercial success.

1892 The diesel engine, used for driving machines, is patented by German engineer Rudolph Diesel.

1894 The *Turbinia*, the first ship with a steam-turbine engine rather than a piston engine, is demonstrated in England.

1937 The first jet engine is demonstrated by the British jet-power pioneer Frank Whittle (b. 1907).

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